SUGARCANE IRRIGATION / MODEL EVALUATION

Four models. One shared test set.

Performance comparison from executed regression notebooks

SIDSS / NIT Patna + NASA POWER | Random seed 42

BEST OBSERVED PERFORMANCE

XGBoost

8.3% lower RMSE than Linear Regression

Lowest MAE and RMSE; highest R-squared.

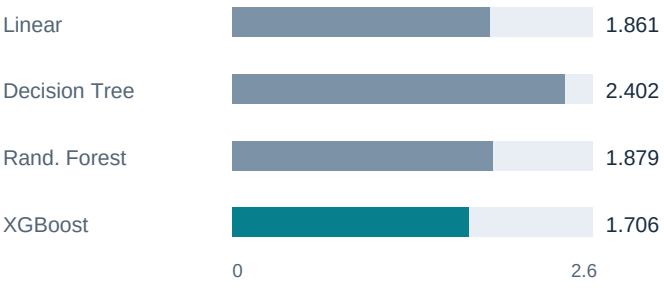
01 Performance at a glance

Errors in billions of configured liters/day (10^9). Lower errors are better.

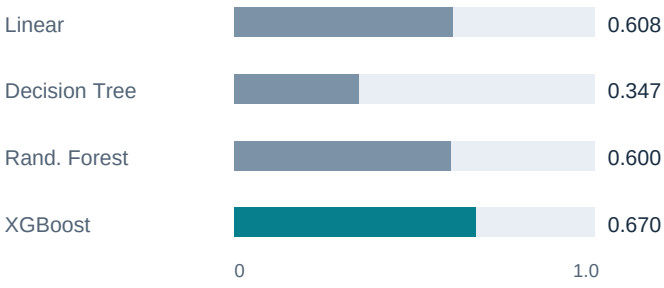
MODEL	MAE / LOWER	RMSE / LOWER	R-SQUARED
Linear Regression	1.2420	1.8610	0.6079
Decision Tree	1.2780	2.4023	0.3467
Random Forest	1.0225	1.8792	0.6002
XGBoost	**0.8029**	**1.7064**	**0.6704**

02 Compare error and explained variation

RMSE (billions of liters/day)



R-squared (higher is better)



03 What the results mean

XGBoost leads on all three metrics on this split. **Random Forest** has lower average absolute error than Linear Regression, but slightly worse RMSE and R-squared, indicating a different pattern of larger errors. **Decision Tree** has the highest errors and lowest R-squared.

Metric guide: MAE = average absolute error. RMSE = error measure that penalizes large misses more strongly. R-squared = explained variation relative to a mean baseline; it is not classification accuracy.

Evaluation: 1,500 rows; identical features and split across models. Initial 60/20/20 split; training + validation combined for final fitting (1,200 rows), with 300 held-out test rows. Saved parameters were reused; no notebook tuning used test outcomes.

Limits: Target magnitudes need unit/provenance validation. NASA weather is repeated across rows because observation dates are absent. Global pre-split clipping and identity/yield inputs limit the evaluation. Selecting a winner by test RMSE requires a separate external test before deployment.